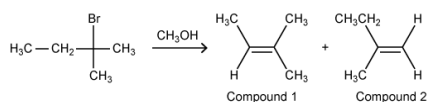


The reaction below was performed under thermodynamic conditions to yield Compound 1 and Compound 2 as a mixture of structural isomers. If Compound 1 is more stable than Compound 2, what can be inferred about the relative amounts of Compound 1 to Compound 2 in the mixture?

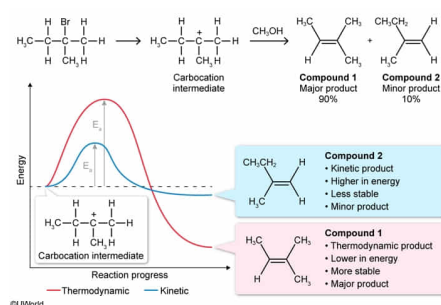


- ✓ ☒ A. There is a greater amount of Compound 1 in the mixture. [84%]
- ☐ B. There is a greater amount of Compound 2 in the mixture. [5%]
- ☐ C. There are equal amounts of Compound 1 and Compound 2 in the mixture. [2%]
- ☐ D. Nothing can be inferred about the relative amounts of the compounds in the mixture. [6%]

Correct

84% answered correctly

Explanation:



Isomers are compounds with the same molecular formula that have a distinguishing structural difference that makes them different compounds with different properties. There are several classifications of isomers, including **constitutional isomers**, **diastereomers**, **enantiomers**, and **cis/trans isomers**. When a reaction is performed, a mixture of isomers can occur with one isomer predominating over the other. In such a case, the reaction conditions dictate which isomer is the major product (such as in the formation of kinetic and thermodynamic **enolates**).

The kinetic product in a mixture requires a lower activation energy to form; therefore, it forms faster than the thermodynamic product but is also higher in energy (less stable). When a reaction is performed at low temperatures (kinetic conditions), insufficient energy is present to supply the larger activation energy required to form the thermodynamic product in high yield. Although the kinetic product still forms faster at higher temperatures (**thermodynamic conditions**), the increased thermal energy enables the kinetic product reaction to occur in reverse (back to the intermediate) and supplies the additional activation energy needed to form the lower energy (**more stable**) **thermodynamic product**, which then predominates.

Compound 1 and Compound 2 are constitutional isomers formed as a mixture in the reaction. Because the reaction was performed under thermodynamic conditions, the more stable isomer will be the major product. Compound 1 is more stable because it contains a more substituted double bond than Compound 2. Therefore, a greater amount of Compound 1 will be in the mixture.

(Choices B and C) Because Compound 2 is less stable than Compound 1 and the reaction was performed under thermodynamic conditions, there will be less of Compound 2 than Compound 1.

(Choice D) Thermodynamic stability of isomers is related to the amount of each isomer in the mixture. Therefore, the relative amounts of each compound can be determined.

Educational objective:

Isomers are different compounds with the same molecular formula, and some reactions produce a mixture of isomers. The major product of a reaction is dependent on the reaction conditions: The more stable product will predominate under thermodynamic conditions, and the less stable product will predominate under kinetic conditions.

Time spent: 0 sec
Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry